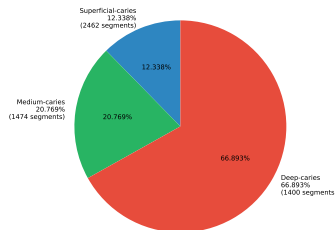


1 Motivation

- ▶ Dental caries is the world's **most common disease**; early X-ray detection is critical.
- ▶ AI is bottlenecked by **scarce annotations** & **class imbalance**.
- ▶ Rare *early* lesions (superficial/medium) are exactly what models miss.

The gap: realistic data for rare classes is missing.

Class Distribution Among Caries Types
(Percentage of Total Caries Pixels & Component Counts)

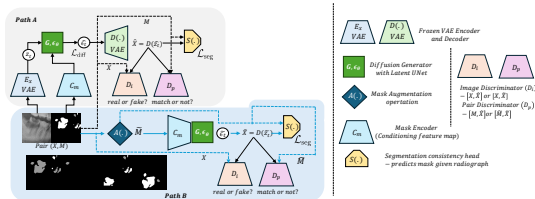


MLUA-DC1000: deep **67%**, medium **21%**,
superficial **12%**.

2 Technical Innovation

CarDiff: segmentation-guided latent diffusion for caries synthesis.

- **Dual-path:** paired + self-supervised (augmented-mask) learning.
- **Anatomy-aware** conditioning: patch-GNN + attention + **FiLM**.
- Adversarial + consistency losses for realism.



Best FID (90.8) & LPIPS (0.48) among 4 baselines.

3

Potential Impact

- ▶ 5× augmentation boosts **segmentation** across 7 nets.
- ▶ Biggest gains on the **under-represented** early lesions.
- ▶ Toward **trustworthy AI diagnosis** under limited data.

+6.5%

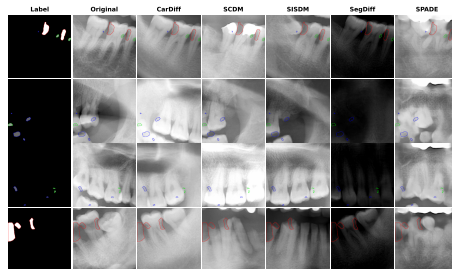
avg. Dice

+25.7%

medium

+16.6%

MC + SC



Synthesized vs. real label–image pairs —
realistic attenuation & sharp lesion
boundaries.